

```
In [76]: from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, r

In [77]: y_true=[0,1,0,0,1,1,0,0,1,1,1,0,1,1,0]

In [78]: y_pred=[1,1,1,0,1,1,0,0,1,0,1,1,0,0,0]

In [79]: confusion_matrix(y_true,y_pred)

Out[79]: array([[4, 3],
               [3, 5]])

In [80]: TN, FP, FN, TP= confusion_matrix(y_true,y_pred).ravel()

In [81]: Acc_lib= accuracy_score(y_true,y_pred)

In [82]: Acc_lib

Out[82]: 0.6

In [83]: Acc_scratch= (TP+TN)/(TP+TN+FP+FN)

In [84]: Acc_scratch

Out[84]: np.float64(0.6)

In [85]: prec_lib= precision_score(y_true,y_pred)

In [86]: prec_lib

Out[86]: 0.625

In [87]: prec_scratch= TP/(TP+FP)

In [88]: prec_scratch

Out[88]: np.float64(0.625)

In [89]: recall_lib= recall_score(y_true,y_pred)

In [90]: recall_lib

Out[90]: 0.625

In [91]: recall_scratch= TP/(TP+FN)

In [92]: recall_scratch

Out[92]: np.float64(0.625)

In [93]: f1_score= f1_score(y_true,y_pred)

In [115... f1_score
```

Out[115... 0.625

Multiclass evaluation

```
In [107... y_true=[1,2,0,1,0,2,1,1,2,2,0,0,1,0,2]
```

```
In [108... y_pred=[0,1,2,1,0,2,1,2,1,0,1,0,1,0,2]
```

```
In [109... confusion_matrix(y_true,y_pred)
```

```
Out[109... array([[3, 1, 1],  
        [1, 3, 1],  
        [1, 2, 2]])
```

```
In [110... Acc_lib= accuracy_score(y_true,y_pred)
```

```
In [111... Acc_lib
```

```
Out[111... 0.5333333333333333
```

```
In [125... precision_multiclass= precision_score(y_true,y_pred,average='weighted')
```

```
In [126... precision_multiclass
```

```
Out[126... 0.5333333333333333
```

```
In [127... recall_multiclass= recall_score(y_true,y_pred,average='macro')
```

```
In [128... recall_multiclass
```

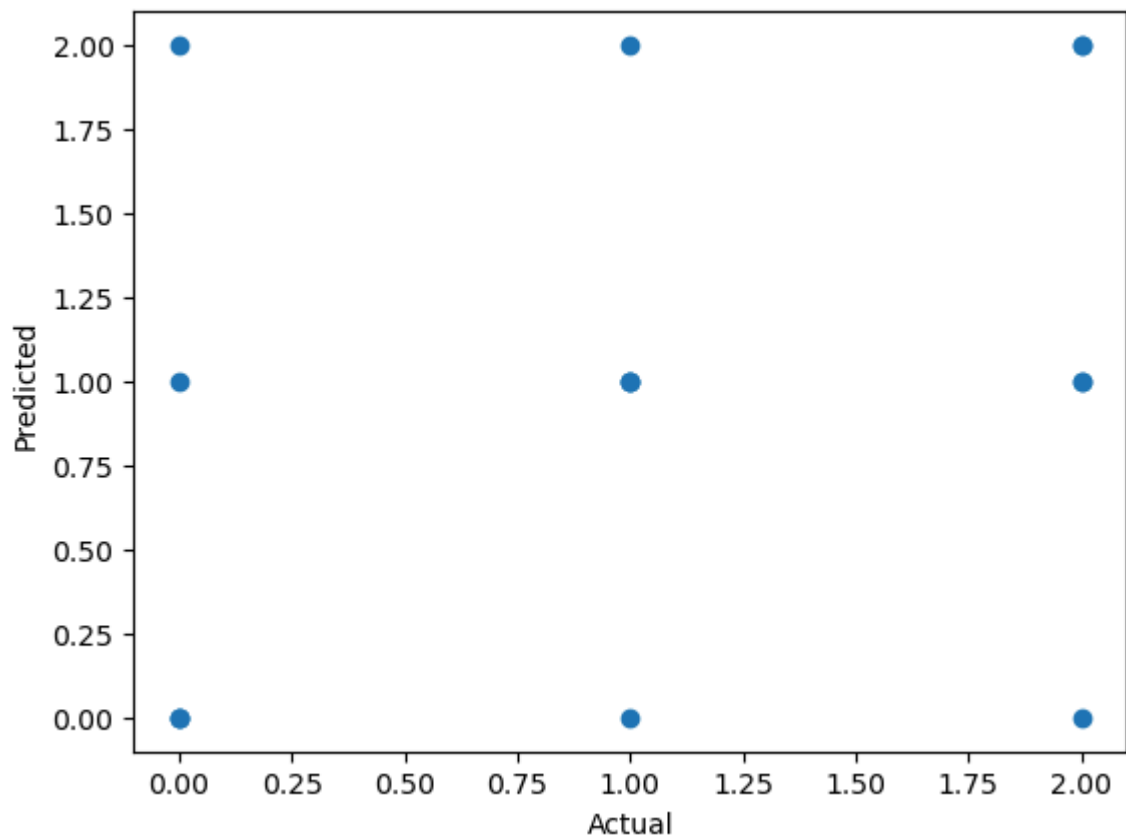
```
Out[128... 0.5333333333333333
```

plotting the graph

```
In [140... import matplotlib.pyplot as plt
```

```
In [141... plt.scatter(y_true,y_pred)  
plt.xlabel('Actual')  
plt.ylabel('Predicted')
```

```
Out[141... Text(0, 0.5, 'Predicted')
```



In []:

In [142... `import pandas as pd`

In [143... `import numpy as np`

load dataset

In [150... `df = pd.read_csv("placement - placement (1).csv")`

In [156... `df`

Out[156...

	cgpa	package
0	6.89	3.26
1	5.12	1.98
2	7.82	3.25
3	7.42	3.67
4	6.94	3.57
...	...	...
195	6.93	2.46
196	5.89	2.57
197	7.21	3.24
198	7.63	3.96
199	6.22	2.33

200 rows × 2 columns

In [157...

```
df.head()
```

Out[157...

	cgpa	package
0	6.89	3.26
1	5.12	1.98
2	7.82	3.25
3	7.42	3.67
4	6.94	3.57

In [158...

```
df.shape
```

Out[158...

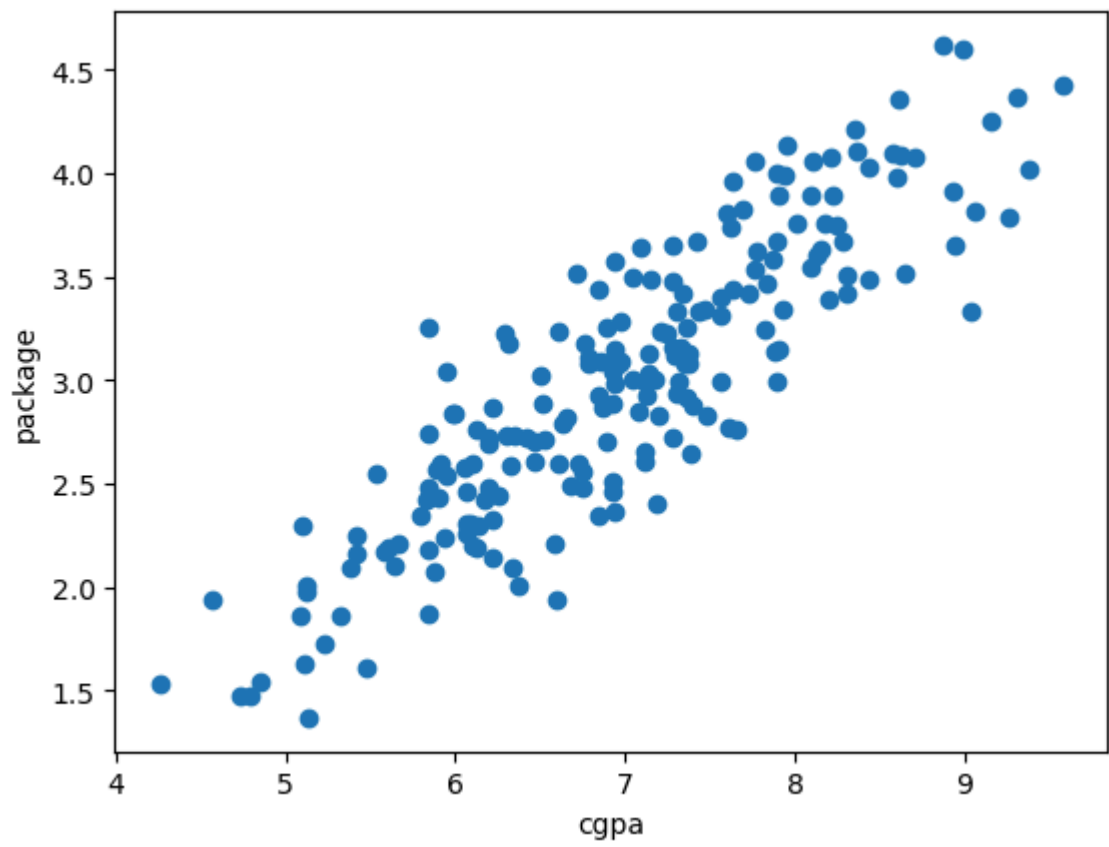
```
(200, 2)
```

In [160...

```
plt.scatter(df['cgpa'], df['package'])  
plt.xlabel('cgpa')  
plt.ylabel('package')
```

Out[160...

```
Text(0, 0.5, 'package')
```



In []: